Aarohi Mathur

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.00
ASHOK HALL HIGHER SEC SCHOOL	Class XII (CBSE)	2020	90.80%
ASHOK HALL HIGHER SEC SCHOOL	Class X (CBSE)	2018	86.00%

POSITIONS OF RESPONSIBILITY

1. **Committee Head, VIMANAZ AARUUSH(1 year)** - Led and coordinated committee activities and team responsibilities during the institute's annual technical fest.
2. **Team Member, SAE INDIA(1 Year)**- Contributed to the design, development and management of activities of the aerospace-focused student team.
3. **Team Member, Rudra (Moon Lander)(2 months)** - Contributed to the student team's lunar-lander project and related technical activities.

SKILLS

Technical Skills:

- Programming - Python (Pandas, NumPy, SciPy, Data science, Matplotlib, Basemap), C, C++, Spyder
- Satellite Data Analysis - SWOT Data Processing & Validation, Geospatial Interpretation.
- Propulsion & Combustion - PEO Based Electrically Controlled Solid Propellants (ECSPs),Propellant preparation and Combustion parameters analysis
- Engineering Software - CATIA, AutoCAD, ANSYS, SolidWorks, NASA CEA

Soft Skills -

Leadership & Team Coordination - Analytical & Critical Thinking - Effective Communication - Research & Scientific Thinking - Project Management - Time Management - Adaptability & Quick Learning

RELEVANT COURSES

- Astromics Workshop - Omspace India Pvt. Ltd. - Observational Astrophysics Workshop - Naxxatra

- Radio Astronomy - NPTEL - Computational Science in Engineering - NPTEL

INTERSHIPS

Space Applications Center, ISRO | *Research Intern* (8 months)

- Processed and analyzed **2 years of SWOT** satellite datasets using Python and Spyder for spatial and oceanographic studies.
- Conducted SWOT satellite data validation and cross-analyzed datasets **across four satellites** to evaluate observation accuracy.

Brahmàstra Aerospace & Defence Pvt. Ltd.| *Intern* (1 month)

- Studied **solid, liquid, and hybrid rocket propulsion**, including propellants, combustion, ignition, and performance.
- Learned designing and analyzing **rocket nozzles and mechanical components** using engineering simulation and problem-solving techniques.

PROJECTS

PEO Based Electrically Controlled Solid Propellant (ECSP) | *M.Tech Thesis*

- Investigated PEO based ECSP formulations, evaluating oxidizer composition and solvent processing on curing characteristics.
- Characterized electrical response, ignition, burning rate and combustion behavior through experimental testing.

Efficient Orbital Maneuvering | *Academic Minor Project*

- Analyzed spacecraft trajectory modification and orbital maneuvering strategies using core principles of orbital mechanics.

Drone Design, Fabrication & Flight Testing | *Workshop*

- Designed, fabricated, and assembled a custom drone, conducting flight tests to evaluate stability and performance.

PVC Manufacturing Process | *Industrial Visit*

- Studied and experienced industrial manufacturing and operational workflows for PVC pipes and window profiles.

HOBBIES AND INTERESTS

- Singing & Music - Badminton - Swimming - Space Exploration & Stargazing - Astronomy

LANGUAGES KNOWN

- English - Hindi